MO89905: Homework 2

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Due: October 27, 2026, 11:59 PM (KST).
Submit by email to the TA: `fairytale@kaist.ac.kr`.

Instructions

Use this homework to understand the methods, your code, and its results. AI coding assistance is allowed: **state which tool you used and for what; there is no penalty for AI use.** You are responsible for checking and explaining your solution.

Submit a PDF report and runnable MATLAB or Python code with a brief run instruction. Acknowledge reused sources and collaboration. Implement the methods using the fixed settings below.

Problem 1 (Exact policy iteration, 30 points)

Consider the normalized lane-keeping model from Section 4.8, with clipping replaced by admissible controls. The lateral error e evolves as

$$e_{k+1} = e_k + u_k + w_k, \qquad \mathcal{X} = \{-4, \dots, 4\}, \qquad \mathcal{U}(e) = \{u \in \{-1, 0, 1\} : |e + u| \le 3\}.$$

Choose u_k before observing w_k. Independently at each step, $\Pr(w_k = -1, 0, 1) = (0.1, 0.6, 0.3)$. A stationary policy μ has cost-to-go

$$J_\mu(e) = \mathbb{E}^\mu \left[\sum_{k=0}^{\infty} \gamma^k g(e_k, \mu(e_k)) \,\middle|\, e_0 = e \right], \qquad g(e, u) = e^2 + 0.2u^2, \quad \gamma = 0.95.$$

Start from $\mu_0(-4) = 1$, $\mu_0(4) = -1$, and $\mu_0(e) = 0$ otherwise. If candidate costs are within 10^{-12} of the minimum, select the first admissible input in the order $-1, 0, 1$.

Part 1 (5 points)

List $\mathcal{U}(e)$ for each state. Explain why all successors remain in $\mathcal{X}$ without clipping or interpolation.

Part 2 (15 points)

Write and implement exact policy evaluation and greedy policy improvement. You may evaluate a policy by solving $(I - \gamma P_\mu) J_\mu = g_\mu$. Repeat until the policy does not change.

Part 3 (10 points)

Report $\mu^*(e)$ and $J^*(e)$ for all nine states, and the number of improvement iterations that change the policy (one count per iteration). Retain the policy sequence for Problem 2's comparison plot.

Problem 2 (MC/TD approximate PI, 70 points)

Use the same model and initial policy. Estimate the current policy's cost-to-go by

$$\widehat{J}_\theta(e) = \phi(e)^\top \theta, \qquad \phi(e) = [1, z, z^2, z^3]^\top, \quad z = e/4.$$

For each MC or TD policy evaluation, collect **100 trajectories of 60 steps**, each starting uniformly from $\mathcal{X}$ and following that policy. Reset $\theta = 0$ for each evaluation. These are segments of a continuing task, with no terminal state. This budget is for this homework; choose and justify your sampling budget separately for the slalom challenge.

MC evaluation. For trajectory j, use its starting state $e_0^{(j)}$ and return

$$G_j = \sum_{k=0}^{59} \gamma^k g(e_k^{(j)}, u_k^{(j)}), \qquad \theta \in \arg\min_{\vartheta} \sum_{j=1}^{100} \big(\phi(e_0^{(j)})^\top \vartheta - G_j\big)^2.$$

Use ordinary least squares (MATLAB backslash or Python `lstsq`), with one return per trajectory start.

TD(0) evaluation. At every sampled transition, update

$$\delta_t = g(e,u) + \gamma \phi(e^+)^\top \theta - \phi(e)^\top \theta, \qquad \theta \leftarrow \theta + \alpha_t \delta_t \phi(e), \qquad \alpha_t = \frac{0.1}{(1 + t/10000)^{0.6}}.$$

Process each trajectory chronologically; count $t = 1, \ldots, 6000$ within each evaluation. Bootstrap at the last step as well. Improve greedily with the known model, using $\widehat{J}_\theta$ in place of J_μ.

Part 1 (25 points)

Using **one random seed**, evaluate the fixed policy μ_0 with MC and TD(0). Plot the two estimated cost-to-go functions and exact J_{μ_0} for all nine states. Report the seed and both RMSEs against J_{μ_0}.

Part 2 (25 points)

For each method, run **five trials with different random seeds** and report them. A trial is a complete four-iteration PI run from μ_0. Use the same five seeds for both methods, setting the seed once at the start of each trial. Iteration $r = 1, \ldots, 4$ evaluates μ_{r-1} with fresh data, then improves it to μ_r. Reuse Part 1 as the first trial's initial evaluation. For each policy, compute

$$d(\mu) = \max_{e \in \mathcal{X}} \big[J_\mu(e) - J^*(e)\big].$$

Plot the mean and sample standard deviation of $d(\mu_r)$ across trials for $r = 0, \ldots, 4$, together with exact PI. For each method, report the final mean gap and how many final policies match $\mu^*(e)$ at *all nine states* (a count from 0 to 5). Exact policy values are diagnostics, never MC/TD targets.

Part 3 (20 points)

Briefly explain: (i) why more 60-step trajectories do not remove MC truncation bias, and why TD still bootstraps at the last step; (ii) whether a cost-to-go estimate with nonzero RMSE can yield an optimal greedy policy; and (iii) for the policy-evaluation and policy-improvement stages, which use samples and which require transition probabilities. Does this small example demonstrate a computational advantage over exact PI?